

Tribhuvan University
Institute of Science and Technology
2076



Bachelor Level / Second Year/ Forth Semester/ Science
Computer Science and Information Technology (CSC 261)
(Artificial Intelligence)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
All figures in the margin indicate full marks.

Attempt all the questions.

Section A

Long Answer Questions

Attempt any Two questions

(2x 10=20)

1. Construct a state space with appropriate heuristics and local costs. Show that Greedy Best first search is not complete for the state space. Also illustrate A* is complete and guarantees solution for the same state space. (10)

2. How resolution algorithm is used in FoPL to infer conclusion?

Consider the facts;

Anyone whom Pugu loves is a star. Any hero who does not rehearse does not act. Anmol is a hero. Any hero who does not work does not rehearse. Anyone who does not act is not a star. Convert above into FoPL and use resolution to infer that "If Anmol does not work, then Pugu does not love Anmol".

(5+5)

3. Define mathematical model of artificial neural network. Discuss how Hebbian learning algorithm can be used to train a neural network. Support your answer with an example. (3+7)

Section-B

Short Answer Questions

Attempt any Eight questions.

(8x5=40)

4. What is AI? How can you define AI from the perspective of thought process? (1+4)
5. Discuss the types of environment where an agent can work on. (5)
6. Illustrate with an example, how uniform cost search algorithm can be used for finding goal in a state space. (5)
7. Define frame. How knowledge is encoded in a frame? Justify with example (2+3)

8. What do you mean by membership of an element in a fuzzy set? Given a domain of discourse $X=\{10,20,30,40,50,60,70\}$, construct a fuzzy set from X. Use your own assumptions for defining membership. (2.5+2.5)
9. Write an algorithm for learning by Genetic Approach. (5)
10. How uncertain knowledge is represented? Given following full joint probability distribution representing probabilities of having different sizes of CD, find the probability that a CD cover has a length of 130mm given the width is 15 mm. (5)

y=width		x=Length →		
↓		129	130	131
	15	0.12	0.42	0.06
	16	0.08	0.28	0.04

11. How the concepts of machine vision are used in Robotics to configure sensors of Robots? (5)
12. How syntactic and semantic analysis is done during natural language processing? Explain with example. (5)